

Automatique

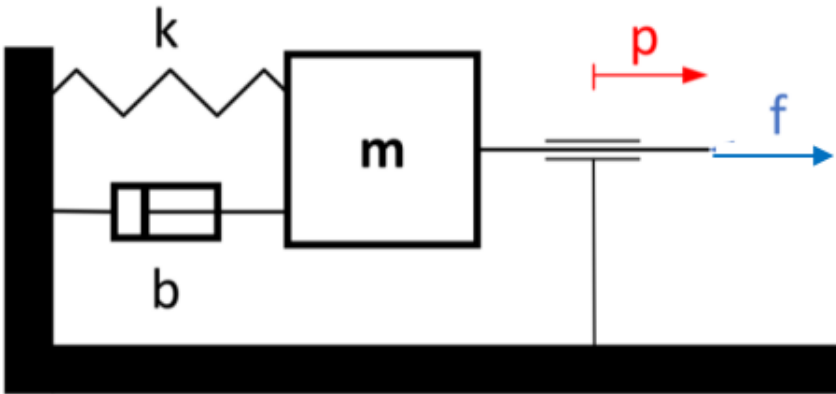
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February 18, 2026

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1 Exercice 1 : Système masse-ressort (matlab)



1.1 A

$$\sum F = ma$$

$$\Leftrightarrow m \cdot \ddot{p}(t) = -k \cdot p(t) - b \cdot \dot{p}(t) + f(t)$$

1.2 B

$$\mathcal{L} \rightarrow m \cdot s^2 \cdot P(s) = -k \cdot P(s) - b \cdot s \cdot P(s) + F(s)$$

$$\Leftrightarrow G(s) = \dots$$

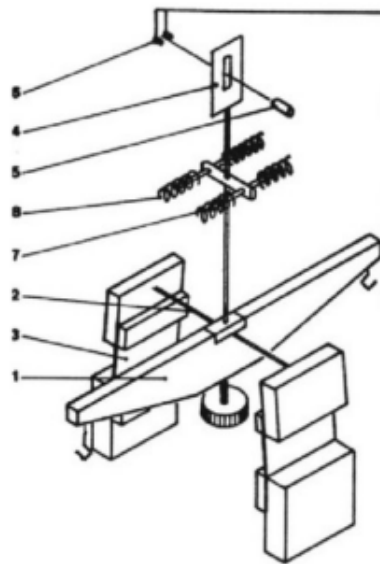
1.3 C

1.4 D

1.5 E

1.6 F

2 Exercice 2 : Système masse-ressort (simulink)



1. Fléau
2. Ruban de torsion
3. Lame de tension du ruban
4. Drapeau
5. Led
6. Eléments photosensibles (phototransistors)
7. Aimants
8. Bobines

2.1 A

2.2 B

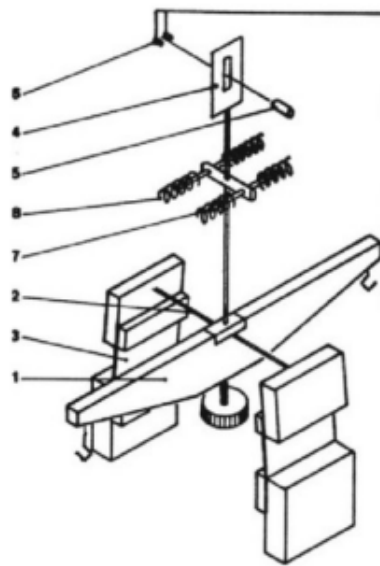
2.3 C

2.4 D

2.5 E

2.6 F

3 Exercice 3 : Balance à fléau



1. Fléau
2. Ruban de torsion
3. Lame de tension du ruban
4. Drapeau
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6. Eléments photosensibles (phototransistors)
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3.1 A

3.2 B

3.3 C

3.4 D

3.5 E

3.6 F